27 Here are the first five terms of an arithmetic sequence.

2 7 12 17 22

The nth term of a different arithmetic sequence is $4n + 15$

The last term of each sequence is the same number.
There are the same number of terms in each sequence.

Find the number of terms in each sequence.

Differences 5; 5n starts at 5, so subtract 3.

1 mark
Nth term 5n − 3

First sequence: 5n − 3.

$$5n - 3 = 4n + 15$$

(−4n) (−4n)

$$n - 3 = 15$$

(+3) (+3)

1 mark
Equal last terms

n = 18 terms; both last terms are 87.

1 mark
18 terms

(Total for Question 27 is 3 marks)